

HOUSTON HOBBY, TX, USA

WMO: 722440

Lat: 29.638N

Lon: 95.282W

Elev: 44

StdP: 14.67

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12918

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.5	36.6	16.5	12.7	43.4	20.9	15.7	46.9	25.1	51.6	22.8	53.4	9.4	0	0.415

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.8	96.3	77.2	94.4	77.3	92.8	77.2	80.5	88.7	80.0	88.2	79.3	87.4	8.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.9	150.1	83.1	77.9	145.4	82.8	77.2	142.0	82.6	44.0	88.6	43.2	87.8	42.7	87.4	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.7	18.6	16.9	28.3	99.9	3.6	3.0	25.7	102.0	23.6	103.8	21.6	105.4	19.0	107.6		
			24.8	82.3	3.1	1.7	22.6	83.5	20.8	84.5	19.0	85.5	16.8	86.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	71.0	55.0	58.7	64.2	70.3	77.5	82.6	84.4	84.8	80.8	72.8	63.5	56.7	(6)
(7)		DBStd	12.59	9.40	9.29	7.86	5.95	4.52	2.98	2.50	2.85	4.25	6.83	8.52	9.32	(7)
(8)		HDD50	138	57	28	7	0	0	0	0	0	0	8	38		(8)
(9)		HDD65	1115	331	213	112	22	1	0	0	0	18	128	290		(9)
(10)		CDD50	7802	210	271	446	611	852	979	1066	1078	925	706	412	246	(10)
(11)		CDD65	3304	19	36	86	183	388	529	601	613	475	259	82	33	(11)
(12)		CDH74	32363	49	113	331	1042	3340	5837	7073	7227	4774	2028	432	117	(12)
(13)		CDH80	12938	0	7	31	176	1088	2506	3226	3370	1888	583	58	5	(13)
(14)	Wind	WSAvg	7.8	9.0	9.2	9.1	9.1	8.3	6.9	6.0	5.8	6.2	7.2	7.9	8.5	(14)
(15)	Precipitation	PrecAvg	52.8	3.9	3.3	2.9	3.3	5.1	7.0	4.4	4.4	5.4	4.6	4.3	3.8	(15)
(16)		PrecMax	83.0	9.6	7.8	7.9	10.5	14.8	23.2	17.2	16.0	15.0	12.5	10.2	9.0	(16)
(17)		PrecMin	26.7	0.2	0.4	0.4	0.1	0.0	0.2	0.1	0.7	0.6	0.0	0.4	0.5	(17)
(18)		PrecStd	15.0	2.3	2.0	1.9	2.7	3.5	5.2	3.4	3.3	3.1	3.5	2.8	2.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.3	80.9	83.2	86.9	92.7	97.5	98.0	99.2	96.5	90.9	84.7	80.8	(19)
(20)			MCWB	66.6	67.3	68.4	70.9	74.6	75.9	77.2	77.3	75.8	75.4	73.1	70.0	(20)
(21)		2%	DB	75.1	77.4	80.3	84.1	90.0	94.0	95.3	96.5	93.2	88.4	81.5	77.2	(21)
(22)			MCWB	66.1	67.2	68.5	70.9	74.8	76.6	77.8	77.7	76.1	74.9	70.9	69.7	(22)
(23)		5%	DB	72.6	74.6	77.9	81.9	88.0	92.0	93.6	94.5	91.1	85.9	79.0	74.5	(23)
(24)			MCWB	65.8	66.5	67.8	70.1	74.7	76.7	77.6	77.8	75.9	73.6	69.9	68.5	(24)
(25)		10%	DB	70.0	72.0	75.1	79.9	85.9	90.3	92.0	92.6	89.2	83.2	76.1	71.8	(25)
(26)			MCWB	64.8	65.9	66.8	69.3	74.3	76.6	77.6	77.7	75.5	71.9	69.0	66.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.2	72.0	73.2	76.1	79.7	81.1	81.4	81.6	80.7	79.1	75.8	73.3	(27)
(28)			MADB	74.0	75.4	78.2	82.2	87.7	89.1	90.1	90.2	87.4	84.4	80.5	76.8	(28)
(29)		2%	WB	69.5	70.3	71.5	74.5	78.0	80.1	80.3	80.4	79.6	77.8	74.2	71.6	(29)
(30)			MADB	72.5	74.1	76.1	79.8	85.2	87.6	89.1	89.2	86.3	83.7	78.6	75.1	(30)
(31)		5%	WB	68.0	69.0	70.5	73.3	77.0	79.2	79.5	79.8	78.6	76.7	72.7	70.1	(31)
(32)			MADB	71.2	72.8	74.8	78.3	83.7	86.6	88.4	88.6	85.2	82.5	77.0	73.6	(32)
(33)		10%	WB	65.9	67.6	69.4	72.3	76.1	78.3	78.9	79.1	78.0	75.5	70.8	67.8	(33)
(34)			MADB	69.3	71.0	73.4	77.1	82.5	85.8	87.4	87.7	84.5	81.0	74.9	71.3	(34)
(35)	Mean Daily Temperature Range	MDBR	17.3	16.6	16.7	16.6	15.7	15.1	15.5	15.8	15.6	17.2	17.6	16.9	(35)	
(36)		5% DB	MCDBR	19.5	17.6	16.9	16.9	17.1	17.2	17.5	18.3	17.3	17.1	17.7	18.3	(36)
(37)			MCWBR	12.8	10.5	8.7	7.8	6.1	4.9	4.8	4.8	5.4	6.9	9.5	12.1	(37)
(38)		5% WB	MCDBR	16.3	14.9	14.3	13.5	14.6	14.5	15.7	16.1	14.3	14.5	15.6	16.6	(38)
(39)			MCWBR	12.8	10.9	9.3	7.8	6.3	5.3	5.1	5.0	5.4	8.0	10.7	12.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.343	0.361	0.375	0.439	0.482	0.472	0.500	0.487	0.458	0.381	0.372	0.352	(40)	
(41)		taud	2.504	2.463	2.414	2.232	2.150	2.228	2.172	2.209	2.258	2.465	2.438	2.485	(41)	
(42)		Ebn at Noon	283	287	289	272	260	261	254	256	260	276	270	273	(42)	
(43)		Edh at Noon	29	32	36	45	49	45	48	45	42	32	30	28	(43)	
(44)	All-Sky Solar Radiation	RadAvg	914	1030	1412	1685	1880	1950	1893	1817	1576	1377	1040	811	(44)	
(45)		RadStd	88	177	120	121	135	156	179	150	166	166	106	97	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+194	+202
(47)	Regional (35 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+200	+175

Nomenclature: See separate page